Definite Integration

Mind map

Meaning
- Signed area
- $F(b) - F(a)$, F continuous

Substitution
- Change limits
- Continuous, one-one
- Reciprocal limits: $x = \frac{1}{t}$

Inverse functions
- $\int_a^b f + \int_c^d g = bd - ac$

P3 splitting
- $|\ |$, $[\ ]$, $\{\ \}$, max/min
- Break at changes

P4 odd / even
- Limits $-a$ to a
- Odd part $= 0$
- $\frac{f}{1+b^x} \to \int_0^a f$

P5 King
- $x \to a + b - x$, add
- $\frac{f}{f+f(a+b-x)} = \frac{b-a}{2}$
- $xf(x)$: $\frac{a+b}{2} \int f$

P6 Queen
- $f(2a - x) = \pm f(x)$
- Lower limit 0
- $\int_0^{\pi/2} \ln \sin x = -\frac{\pi}{2} \ln 2$

P7 Periodic
- $\int_a^{a+nT} f = n \int_0^T f$
- Remove whole periods

Leibnitz
- $f(h)h' - f(g)g'$
- Take x out first
- $\frac{0}{0}$: L'Hospital

Determination of function
- Constant limits: $= k$
- Variable limits: differentiate

Limit of sum
- $\frac{r}{n} \to x$, $\frac{1}{n} \to dx$
- Product: take ln
- Left $< \int <$ right ($f \uparrow$)

Estimation
- $m(b - a) \leq \int \leq M(b - a)$
- Concavity: tangent / chord
- Compare functions

Wallis
- $\int_0^{\pi/2} \sin^n x \cos^m x$
- $K = \frac{\pi}{2}$ if both even

Reduction
- Parts: I_n to I_{n-1}, I_{n-2}
- $I_n + I_{n-2} = \frac{1}{n-1}$ (tan)